Sequences — problem 9.48

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The sequence a_n is defined by $a_0 = 0$, $|a_2| = |a_1 + 1|, \ldots, |a_n| = |a_{n-1} + 1|$. Prove that

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq -\frac{1}{2}.$$